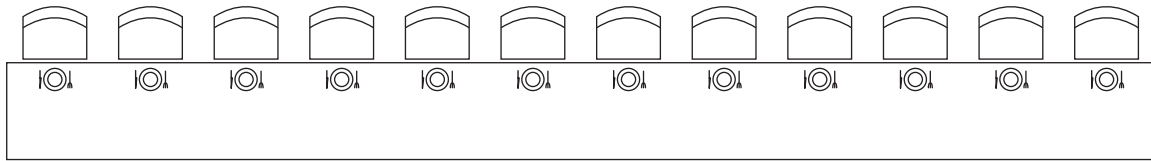


7. [Maximum mark: 6]

A self-service sushi restaurant has a row of 12 available seats, as shown in the following diagram.



Anvi, Vanya and Parita decide to go to the restaurant for lunch.

- (a) Find the number of possible ways that they can be seated in this row, if they decide **not** to sit together as a group of 3. [3]

The next day, Anvi, Vanya and Parita are joined by 3 additional people in the same restaurant, and they sit in the same row of 12 available seats. Anvi, Vanya and Parita now decide to sit next to each other as a group of 3.

- (b) Find the number of possible ways that these 6 people can be seated. [3]

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